

Total No. of Questions : 8]

PE2474

SEAT No. :

[Total No. of Pages : 2

[6584]-456

B.E. (Computer Engineering)

**HONORS IN DATA SCIENCE : MACHINE LEARNING AND
DATA SCIENCE**

(2019 Pattern) (Semester - VII) (410501)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.

- Q1)** a) What is the role of dendrograms in choosing number clusters in Hierarchical clustering? [8]
- b) Explain in detail KNN technique with example. [9]

OR

- Q2)** a) What is clustering? Explain Agglomerative Hierarchical Clustering technique. [8]
- b) What is K-Means algorithm? Explain in detail. [9]

- Q3)** a) What is Activation Function? Explain its types in detail. [9]
- b) What is Perceptron? Explain the process of training a perceptron? [8]

OR

- Q4)** a) What is a Neural network? Explain Back propagation Algorithm in detail. [9]
- b) Explain the following concept in detail. [8]
- i) Generalized Delta Learning Rule.
 - ii) Multilayer Perceptron's

P.T.O.

- Q5)** a) Explain the concept of Long - short Term memory with example. [9]
b) Describe Gradient descent optimization in depth. [9]

OR

- Q6)** a) Explain CNN algorithm and its architecture with a diagram. [9]
b) What is Recursive Neural Network? Explain Recurrent Neural Network with example. [9]

- Q7)** a) Describe the concept Text Preprocessing with suitable example. [9]
b) Explain Topic modeling algorithms in detail. [9]

OR

- Q8)** a) Explain in detail. [9]
i) Tokenization
ii) Feature selection and Extraction.
b) What is document representation in text preprocessing. Explain Text Similarity measure in detail. [9]

